Toward Simple Best-of-Both-Worlds Algorithms: Insights from Follow-the-Perturbed-Leader

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Outline & objective

- Focus on intuition rather than introducing the results in detail.
  - ∵ anyone can understand math by reading the paper.
- Try to explain some fundamental things that can be applied to
  - other frameworks, researches

Environmental assumptions in bandits

- Given K arms, the environment sets loss vector $\ell_t \in [0,1]^K$.
- Two primary environments that determines loss sequences.
- Stochastic setting:
  - Loss vector $\ell_{t,i} \overset{\text{i.i.d.}}{\sim} \mathcal{P}_i$ for unknown fixed distribution $\mathcal{P}_i$ over $[0,1]$.
  - Lower bound: $\mathrm{Reg}(T) \geq \Omega(\sum_{i \neq i^*} \log T / \Delta_i)$ where $\mu_i = \mathbb{E}_{X \sim \mathcal{P}_i}[X]$, $\Delta_i = \mu_i - \mu_{i^*}$, and $i^* = \min_{j \in [K]} \mu_j$.
- Adversarial setting:
  - Loss vector is determined by an (adaptive) adversary.
  - Lower bound: $\mathrm{Reg}(T) \geq \Omega(\sqrt{KT})$.

Environmental assumptions in bandits

Question.

How can we be certain whether the environment is stochastic, adversarial, or somewhere in between?



- Intermediate: non-stationary, AwSC.
- Adversarial regime with self-bounding constraints (Δ, C, T) (AwSC): $\mathrm{Reg}(T) \geq \sum_{t=1}^{T} E[\Delta_{i_t}] - C$, where i_t is played arm by a policy.

Environmental assumptions in bandits

Question.

How can we be certain whether the environment is stochastic, adversarial, or somewhere in between?



We want a policy that adapts automatically to those environments without knowing the environment.

- We call such a policy Best-of-Both-Worlds (BOBW) policy.

Test and Switch

Intuition

- Run stochastic policy at first.
- Employ statistical tests to detect non-stochastic behavior.
- Switch to some robust policy for adversarial setting.

- Earlier BOBW policies adopt this approach, including the first policy by Bubeck and Slivkins (2012).
- Auer and Chiang (2016) in multi-armed bandits, Lee+ (2021) in linear bandits, Kong+ (2022) in graph bandits.

Test and Switch

Intuition

- Run stochastic policy at first.
- Employ statistical tests to detect non-stochastic behavior.
- Switch to some robust policy for adversarial setting.

Disadvantages

- Algorithmic complexity:
  two separate policies with a switching mechanism.
- Risk of incorrect or delayed switching.
- Potentially loose bounds in intermediate regimes.

- In intermediate regimes, the desired guarantee should interpolate between the stochastic and adversarial bounds.

Follow-the-Regularized-Leader (FTRL)

Intuition

- Add penalty term to loss estimates.
- Play an arm i_t according to arm-selection distribution w_t

$$w_t \in \arg\min_{p \in \mathcal{S}^K} \left\{ \left\langle w, \hat{L}_t \right\rangle + \psi(w)/\eta_t \right\}.$$

- $\mathcal{S}^K$: $K-1$ dimensional probability simplex.
- $\hat{L}_t = \sum_{s=1}^{t-1} \hat{\ell}_t \in \mathbb{R}^K$: cumulative loss estimator.
- $\psi : \mathcal{S}^K \to \mathbb{R}_{\geq 0}$: regularization function.
  - e.g., ℓ_1 norm, ℓ_2 norm, Shannon entropy.
- η_t: learning rates, e.g., $1/\sqrt{T}$, $1/\sqrt{t}$.

Follow-the-Regularized-Leader (FTRL)

Advantages

- Explicit characterization of the distribution w_t.
- Flexible control of w_t through regularizer design.
- Systematic analysis using convex duality and Bregman divergence.

- Very successful framework in BOBW literature.
  - Multi-armed bandits, linear contextual bandits, graph bandits, partial monitoring, heavy-tailed bandits, combinatorial semi-bandit, decoupled bandits, just to name a few.

Disadvantages

- Computational complexity:
  need to solve optimization problem at every round.

Follow-the-Perturbed-Leader (FTPL)

Intuition

- Add random noise to loss estimates.
- Play an arm i_t that minimizes perturbed loss estimates

$$i_t \in \arg\min_{i \in [K]} \left\{ \hat{L}_{t,i} - r_{t,i}/\eta_t \right\}.$$

- $r_t \sim \mathcal{D}$: a random noise vector (perturbation).
- $\mathcal{D}$: a perturbation distributions.
  - e.g., i.i.d. Gaussian distributions $\rightarrow r_{t,i} \sim \mathrm{Normal}(0,1)$ for all t, i.

Follow-the-Perturbed-Leader (FTPL)

Advantages

- Requires only a linear optimization.
- Simple randomized decision rule and easy implementation.

- Computationally attractive counterpart of FTRL.

Disadvantages

- Difficulty in analysis:
  the formulation of w_t does not have closed form in most cases.

- Why w_t is important in analysis?
  - Expected regret naturally reflect how the policy plays the arm i_t, i.e., w_t.

Previous results in FTPL

- Focus on i.i.d. perturbations with distribution function F.
- $w_{t,i} = \Pr\left(i_t = i \middle| \hat{L}_t\right)$:
  probability that FTPL plays arm i given cumulative loss estimates $\hat{L}_t$.

$$w_{t,i} = \int_{-\infty}^{\infty} \prod_{j \neq i} F(z + \eta_t \underline{\hat{L}}_{t,j}) \mathrm{d}F(z + \eta_t \underline{\hat{L}}_{t,i}),$$

  where $\underline{\hat{L}}_{t,i} = \hat{L}_t - \min_{j \in [K]} \hat{L}_{t,j}$.
- When F is Gumbel distribution, $w_{t,i}$ becomes logit model.
  - $w_{t,i} \propto \exp\left(-\eta_t \underline{\hat{L}}_{t,i}\right)$.
- In general, no closed form can be obtained.

Previous results in FTPL

Big difference

$\sqrt{KT}$ guarantee of FTPL was unknown until 2023.
The optimality of FTRL was obtained by Audibert and Bubeck (2009).

- The use of $1/2$-Tsallis entropy as a regularizer.
  - $\psi(w) \approx -\sum_{i\in[K]} \sqrt{w_i}$.
  - $\mathcal{O}(\sqrt{KT})$ adversarial regret and $\mathcal{O}(\log(T))$ stochastic regret (Zimmert and Seldin, 2021).

Previous results in FTPL

Question.

What kind of distribution can achieve the optimal results?

- The distribution function F control the behavior of w_t.

The choice of perturbation distributions

- FTPL plays an arm with the *smallest perturbed loss*.
  - Let arm i have 3rd smallest loss among 10 arms.
  - Arm i can be selected only if $r_{t,i}$ is larger than
    - perturbation of arm with 1st smallest loss
    - perturbation of arm with 2nd smallest loss

Observation

The arm-selection probability is determined by block maxima of perturbations!

Extreme value theory (EVT)

- EVT studies the behavior of block maxima of random variables.

Informal Theorem of Fisher and Tippet, (1928); Gnedenko, (1943)

The behavior of maximum can be classified into three types.

- Gumbel-type: exponential-like tail, e.g., Gumbel, Gaussian.
- Fréchet-type: heavy-tail, e.g., Pareto, Fréchet.
- Weibull-type: bounded tail, e.g., uniform, Beta.

Previous results

- Gumbel-type: only near-optimal results, $\tilde{\mathcal{O}}(\sqrt{KT})$.
  - Exponential, Gaussian, Gumbel (a.k.a. Exp3).
  - Poland (2005), Abernethy+ (2015), Li and Tewari (2018), Kim and Tewari (2019).
- Weibull-type: known to be suboptimal (Kim and Tewari, 2019).

Previous results

Conjecture by Kim and Tewari (2019)

If FTPL is optimal, then such F will be Fréchet-type.

Kim, Tewari, On the optimality of perturbations in stochastic and adversarial multi-armed bandit problems, NeurIPS 2019

Regularization and perturbation across fields

Regularization

$$w(u) = \arg\min_{w \in \mathcal{S}^K} \{\langle u, w \rangle + \psi(w)\}$$

Random perturbation

$$I(u, r) = \arg\min_{i \in [K]} \{u_i - r_i\}.$$

Field	**Regularization**	**Random perturbation**
Online learning	FTRL	FTPL
Game theory	Regularized fictitious play	Stochastic fictitious play
Discrete choice	Representative agent model	Additive random utility model

The same mechanisms appear under different names.
(NB; the objective is different)

A bridge from discrete choice theory

Regularization

$$w(u) = \arg\min_{w \in \mathcal{S}^K} \{\langle u, w \rangle + \psi(w)\}$$

Random perturbation

$$I(u, r) = \arg\min_{i \in [K]} \{u_i - r_i\}.$$

- Two policies are the same if
$$w_i(u) = \Pr(I(u, r) = i),\ \forall u \in \mathbb{R}^K, i \in [K].$$
- FTRL with Shannon entropy ($\psi(w) = -\sum_i w_i \log w_i$)
  = FTPL with Gumbel perturbation ($f(r) \approx e^{-x+e^{-x}}$)
  = Exp3 ($w(u) \propto \exp(-u)$)

A bridge from discrete choice theory

Results in discrete choice theory

$$\text{FTPL} \subsetneq \text{FTRL}$$

Every FTPL can be represented by FTRL.

The converse does not hold in general.

- Given distribution F, there exists a corresponding ψ.

A bridge from discrete choice theory

- FTRL with Tsallis entropy is BOBW with optimal $\mathcal{O}(\sqrt{KT})$ regret.

Question.

Are there some distributions F satisfying

$$\text{FTRL with Tsallis entropy} \to \text{FTPL with } F?$$

A bridge from discrete choice theory

Negative results by Kim and Tewari (2019)

When $K \geq 4$, there is no such F.

A bridge from discrete choice theory

Negative results by Kim and Tewari (2019)

When $K \geq 4$, there is no such F.

(Informal) Positive results by Abernethy+ (2014, 2016)

When $K = 2$ (one-dimensional optimization), FTPL $=$ FTRL.

Kim, Tewari, On the optimality of perturbations in stochastic and adversarial multi-armed bandit problems, NeurIPS 2019
Abernethy, Lee, Sinha, Tewari, Online linear optimization via smoothing, COLT 2014
Abernethy, Lee, Tewari, Perturbation techniques in online learning and optimization, 2016.

A bridge from discrete choice theory

Negative results by Kim and Tewari (2019)

When $K \geq 4$, there is no such F.

(Informal) Positive results by Kim and Tewari (2019)

When $K = 2$, FTRL with Tsallis entropy can be seen as FTPL with a Fréchet-type perturbation.

- The difference between two perturbations follows Fréchet-type.

First optimal FTPL for bandits

Honda+ (2023)

- FTPL with Fréchet perturbations of shape $\alpha = 2$.
- Optimal adversarial regret

$$\mathrm{Reg}(T) = \mathcal{O}(\sqrt{KT}).$$

- Logarithmic stochastic regret

$$\mathrm{Reg}(T) = \sum_{i \neq i^*} \mathcal{O}\left(\frac{\log T}{\Delta_i}\right).$$

- The same policy achieves both bounds without testing or switching.

First BOBW guarantee for FTPL in multi-armed bandits

Honda, Ito, Tsuchiya, Follow-the-Perturbed-Leader achieves best-of-both-worlds for bandit problems, ALT 2023.

Analytical idea

In BOBW literature, a standard analysis technique requires (Bubeck, 2019)

$$-\frac{\partial_i w_i}{w_i} \lesssim \sqrt{w_i}. \qquad \text{(sufficient condition)}$$

- FTRL with $1/2$-Tsallis entropy satisfies this condition.

Results in Honda+ (2023)

$-\frac{\partial_i w_i}{w_i} \lesssim \frac{1}{\sqrt{\sigma_i}}$, where σ_i is the rank of loss estimates.

- Smallest loss $\to \sigma = 1$.
- Note that $w_i \leq \frac{1}{\sigma_i}$ must hold.
- A loose upper bound without knowing w_i in FTPL.

Beyond one distribution

Remaining question

Is the exact Fréchet form with $\alpha = 2$ essential?

- $F(x) = \mathbb{1}[x \geq 0] e^{-x^{-2}}$.
- $f(x) = 2x^{-3} e^{-x^{-2}}$ for $x \geq 0$.

From one distribution to a tail class

First research question

Which properties of F are essential for optimal guarantee?

- From one specific distribution to a general tail class.
- Sufficient conditions?

From one distribution to a tail class (COLT2024)

Extreme value theory characterizes Fréchet-type tails by

$$1 - F(x) = x^{-\alpha} S_F(x),\ x \geq 0.$$

where $S_F(x)$ is a slowly varying factor that does not change the power-law exponent α, e.g., $S_F(x) = 1$ for Pareto distribution.

Our first result (Lee+, 2024)

Under some mild conditions,

FTPL with Fréchet-type perturbation achieves BOBW guarantee.

- Examples: Fréchet, (Generalized) Pareto, Student-t, Snedecor's F.
- Resolves the conjecture in a comprehensive way.

Lee, Honda, Ito, and Oh, Follow-the-perturbed-leader with Fréchet-type tail distributions: Optimality in adversarial bandits and best-of-both-worlds, COLT 2024

Remaining research questions

Our result

FTPL with Fréchet-type of tail index $\alpha = 2$ is BOBW in MABs.

1. **Exact correspondence**

   Which perturbation model reproduces the Tsallis FTRL?

   Possibly beyond i.i.d. additive perturbations.

2. **Beyond MABs**

   FTRL has achieved BOBW guarantees in many learning problems.

   Which of its design principles can be transferred to FTPL?

1. From Tsallis entropy to Fréchet-type (NeurIPS2025)

Kim and Tewari (2019)

When $K = 2$,

β-Tsallis entropy $\to r_1 - r_2$ follows Fréchet-type with $\alpha = \frac{1}{1-\beta}$.

- A correlated perturbation representation always exists.
- The existence of an i.i.d. representation remains open.

Our refinements (Lee+, 2025)

- If it exists, its marginal distribution must be fully supported on $\mathbb{R}$.

Lee, Honda, Ito, and Oh, Revisiting Follow-the-perturbed-leader with unbounded perturbations in bandit problems, NeurIPS 2025

1. From Tsallis entropy to Fréchet-type (NeurIPS2025)

- We cannot obtain the exact formulation of such distributions.
- But we have characteristic function.

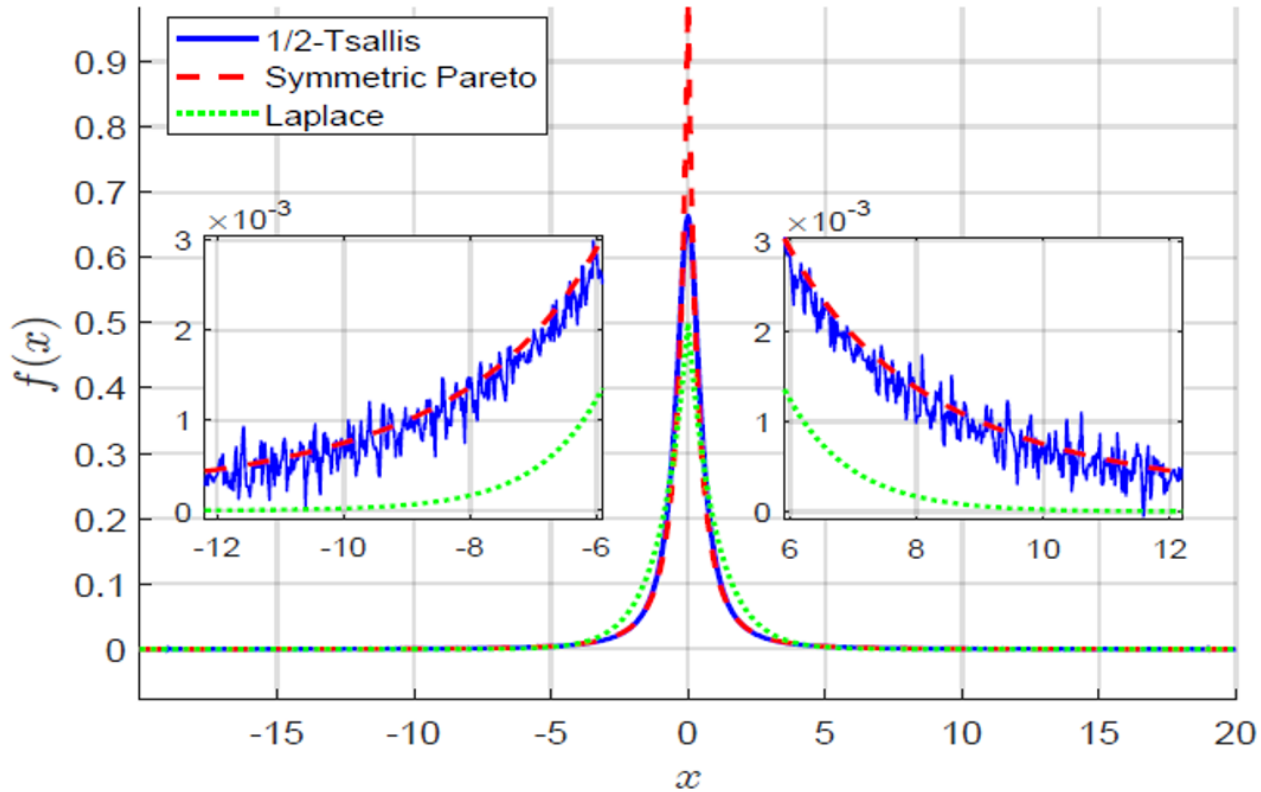


Figure: Inverse Fourier transform of characteristic function.

1. From Tsallis entropy to Fréchet-type (NeurIPS2025)

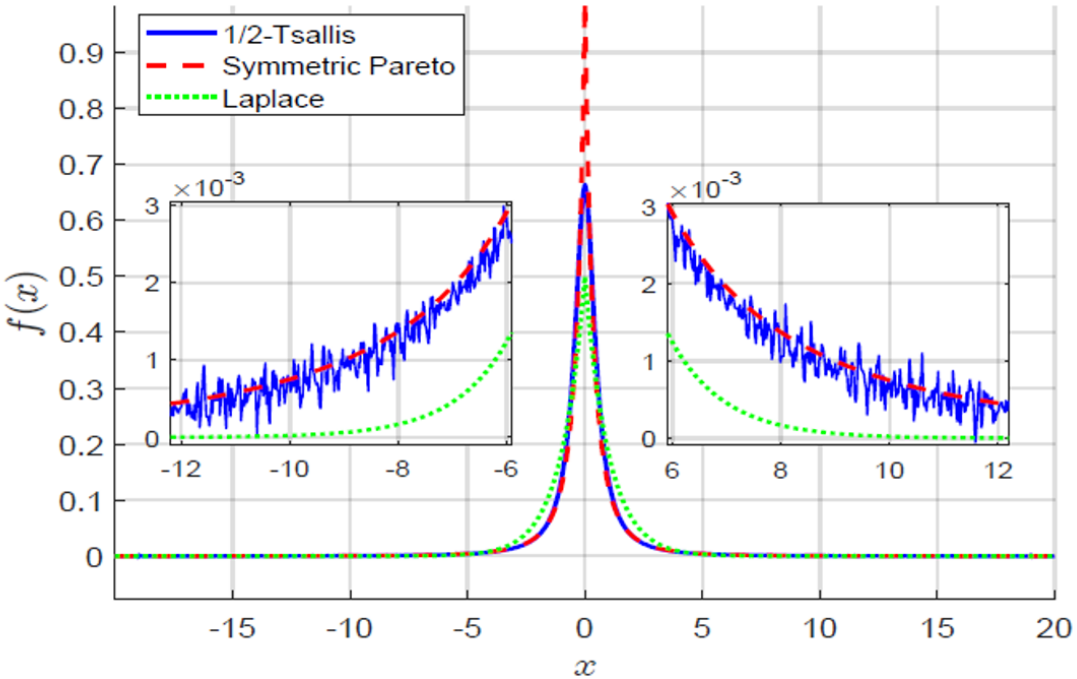


Figure: Inverse Fourier transform of characteristic function.

Our conjecture

There exists a symmetric distribution that reproduces FTRL with $1/2$-Tsallis entropy.

1. From Tsallis entropy to Fréchet-type (NeurIPS2025)

Our result in COLT2024

Applicable only for semi-infinite support distributions.

Remaining question

- Symmetric perturbation can achieve optimal results?
- Such correspondence is restricted only to $K = 2$.
- There is no such perturbations for $K \geq 4$. (Kim and Tewari, 2019)

1. From Tsallis entropy to Fréchet-type (NeurIPS2025)

Positive results for two-armed bandits

FTPL with symmetric Pareto with shape $\alpha = 2$ achieves BOBW.

Negative results for $K \geq 3$

For $\lambda = (0, c, \ldots, c)$ with $c \geq 2\sqrt{K}$, symmetric Fréchet-type perturbations with index $\alpha = 2$ can satisfy

$$-\frac{\partial_i w_i(\lambda)}{w_i(\lambda)^{3/2}} \geq \tilde{\Omega}\left(\frac{c}{K\sqrt{K}}\right), \qquad i \neq 1.$$

This violates the standard sufficient condition for BOBW analysis.

- NB; it does not prove the impossibility of BOBW.

2. Beyond multi-armed bandits

Our approach so far

We have transferred successful FTRL designs to FTPL.

- Several FTRL policies achieve BOBW guarantees in various problems.

Can we utilize their approaches?

2. Beyond multi-armed bandits

FTRL

$$w_t = \arg\min_{w \in \mathcal{S}^K} \left\langle \hat{L}_t, w \right\rangle + \psi(w)/\eta_t$$

FTPL

$$i_t \in \arg\min_{i \in [K]} \left\{ \hat{L}_{t,i} - r_{t,i}/\eta_t \right\}.$$

Three main design components.

1. Regularizer / Perturbation: the main design component.
2. Learning rate η_t: control the adaptivity of policy.
3. Loss estimator $\hat{\ell}_t$: partial feedback of bandits.

2. Main design components in FTRL

Main focus of BOBW policy design

1. **Regularizer design**
   Hybrid regularizers combine properties suited to different settings.
   - e.g., sum of two different regularizer functions.
   - $\psi(w) \approx -\sum_i \sqrt{w_i} + (1 - x_i)\log(1 - x_i)$ in Zimmert+ (2019).
2. **Learning-rate design**
   Adaptive learning rates provide problem-dependent guarantees.
   - Update η_t as a function of w_t.

Usually kept standard

3. **Loss estimation**
   The standard importance-weighted (IW) estimator

$$\ell_t = \mathbb{1}[i_t = i]\ell_{t,i}/w_{t,i}.$$

   is usually sufficient since w_t is explicitly available.

Most advances have focused on the first two components.

2-1. Hybrid perturbations

First question

Can we design a kind of hybrid perturbations for FTPL?

1. **Additive construction**
   Combine perturbations from different distribution families.

$$r = r^{(1)} \pm r^{(2)}, \text{ where } r^{(1)} \sim N(0,1),\ r^{(2)} \sim \mathrm{Par}(2).$$

2. **Asymmetric construction**
   Use different distributions for the left and right tails.

$$f(x) = \begin{cases} e^{2x}, \text{ (Laplace)} & \text{if } x < 0, \\ 1/(x+1)^3, \text{ (Pareto)} & \text{if } x \geq 0. \end{cases}$$

2-1. Hybrid perturbation design (NeurIPS2025)

Optimality of asymmetric perturbations (Lee+, 2025)

FTPL achieves a BOBW guarantee when the right tail is Fréchet-type with index $\alpha = 2$ and the left tail is lighter with index $\alpha' \geq \alpha + 2$.

- This class includes the Laplace-Pareto perturbation.
- The two tails may also be Fréchet-type with different indices.

Lee, Honda, Ito, and Oh, Revisiting Follow-the-perturbed-leader with unbounded perturbations in bandit problems, NeurIPS 2025

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- This class includes the Laplace-Pareto perturbation.
- The two tails may also be Fréchet-type with different indices.

Tail asymmetry may be more fundamental than hybridization itself.

- Symmetric perturbation corresponds to Tsallis entropy.
- A standard Pareto distribution is already asymmetric.

Lee, Honda, Ito, and Oh, Revisiting Follow-the-perturbed-leader with unbounded perturbations in bandit problems, NeurIPS 2025

2-1. m-set combinatorial semi-bandits

FTRL with hybrid regularizer (Zimmert+, 2019)

$$\psi(w) \approx -\sum_{i=1}^{K} \sqrt{w_i} + (1 - x_i)\log(1 - x_i)$$

achieves BOBW guarantee with $\mathcal{O}(\sqrt{mKT})$ adversarial regret.

Computational motivation

FTRL: Regularized optimization over $\left\{w \in [0,1]^K : \sum_i w_i = m\right\}$
FTPL: Play arms with m smallest perturbed losses (top-m selection).

Open question

Can computationally attractive FTPL achieve the same guarantees?

Zimmert, Luo, Wei, Beating stochastic and adversarial semi-bandits optimally and simultaneously, ICML 2019

2-1. Previous results in FTPL

Neu and Bartók (2016)

FTPL with exponential distribution achieves $\mathcal{O}(m\sqrt{KT\log(K/m)})$ regret with computational cost of $\mathcal{O}(K^2)$.

Zhan+ (2025)

FTPL with shape-2 Fréchet perturbations achieves BOBW guarantee with $\mathcal{O}(\sqrt{mKT\log K} + \sqrt{m^{8/3}T})$ adversarial regret.

The optimal adversarial guarantee remained open.

Neu, Bartók, Importance weighting without importance weights: An efficient algorithm for combinatorial semi-bandits, JMLR 2016.
Zhan, Xin, Sun, Zhang, Follow-the-perturbed-leader nearly achieves best-of-both-worlds for the m-set semi-bandit problems, NeurIPS 2025

2-1. Optimal FTPL for m-set semi-bandits (Preprint)

Our result (Chen+ 2026)

For every $\alpha > 1$, FTPL with either Fréchet or Pareto perturbations obtains the adversarial regret of order

$$\mathrm{Reg}(T) = \mathcal{O}(\sqrt{mKT}).$$

- The bound matches the minimax lower bound.
- If $\alpha = 2$, FTPL also achieves logarithmic stochastic regret (BOBW).
- We further reduce the computational cost to $\mathcal{O}(mK(\log(K/m) + 1))$.

A single Fréchet-type perturbation can achieve the same guarantee to FTRL with hybrid regularizers.

Chen, Lee, Kim, Honda, A further efficient algorithm with best-of-both-worlds guarantees for m-set semi-bandit problem, arXiv 2026 (under review).

2-1. Quick recap

Framework	Design choice	Learning rate	Loss estimator
Recent FTRL	Hybrid regularizer **Main focus**	adaptive to w_t **Main focus**	IW estimator Standard choice
Our FTPL	Pareto perturbation Kept simple	Simple $\eta_t = 1/\sqrt{t}$	IW estimator? Standard choice

A single Fréchet-type perturbation can achieve the same guarantee to FTRL with hybrid regularizers.

2-2. Dependence on the $1/2$-Tsallis entropy

Canonical optimal BOBW result

For multi-armed bandits, the standard optimal result relies on the $1/2$-Tsallis regularizer and Fréchet-type with index $\alpha = 2$.

How can we move beyond one fixed design?

- Optimal adversarial regret is obtained for any β-Tsallis entropy ($\beta \in (0, 1)$) and Fréchet-type with $\alpha > 1$.
- More complex problems may require additional flexibility.

2-2. Flexible designs in FTRL

Recent FTRL approaches combine

- **Hybrid regularizers**
  β-Tsallis entropy with an additional regularizer, such as a log-barrier or $(1-\beta)$-Tsallis entropy.
- **Adaptive learning rates**
  Beyond the simple schedule $\eta_t = 1/\sqrt{t}$, update η_{t+1} using the selection probabilities w_t.

The BOBW guarantee extends to

- general β-Tsallis entropy in multi-armed bandits (Jin+ 2023, Ito+ 2024)
- difficult problems with $\Theta(T^{2/3})$ minimax bound (Tsuchiya and Ito 2024)

Jin, Liu, Luo, Improved best-of-both-worlds guarantees for multi-armed bandits: FTRL with general regularizers and multiple optimal arms, NeurIPS 2023
Ito, Tsuchiya, Honda. Adaptive learning rate for follow-the-regularized-leader: Competitive analysis and best-of-both-worlds, COLT 2024
Tsuchiya and Ito, A simple and adaptive learning rate for FTRL in online learning with minimax regret of $\Theta(T^{2/3})$ and its application to best-of-both-worlds, NeurIPS 2024

2-2. Flexible designs in FTRL

Recent FTRL approaches combine

- **Hybrid regularizers**
  to control the behavior of w_t.
- **Adaptive learning rates**
  to adapt to the problem's difficulty.

These adaptive updates rely on explicit access to w_t.

2-2. The missing probability in FTPL

FTPL can easily sample an action

$$i_t = \arg\min_{i \in [K]} \left\{ \hat{L}_{t,i} - r_{t,i}/\eta_t \right\}.$$

However, its selection probability

$$w_{t,i} = \Pr\left(i_t = i \middle| \hat{L}_t \right)$$

is generally difficult to compute.

- Sampling from w_t is easy.
- Evaluating w_t explicitly can be intractable.

Next research question

Can FTPL achieve probability-dependent adaptivity without computing w_t?

2-2. The use of surrogate probability (COLT2026)

Yes, with surrogate probability.

Key observation

Previous FTPL analyses control $w_{t,i}$ using **computable bounds** based on the rank $\sigma_{t,i}$ and estimated loss gaps.

Lee, Honda, Ito, Kim, Adaptive learning rates with surrogate probability for follow-the-perturbed-leader, COLT 2026

2-2. The use of surrogate probability (COLT2026)

Key observation

Previous FTPL analyses control $w_{t,i}$ using **computable bounds** based on the rank $\sigma_{t,i}$ and estimated loss gaps.

Our idea

Use these bounds as surrogate probabilities to update η_t.

- No explicit computation of w_t
- Only quantities already available to FTPL

Lee, Honda, Ito, Kim, Adaptive learning rates with surrogate probability for follow-the-perturbed-leader, COLT 2026

2-2. The use of surrogate probability (COLT2026)

Main results

- BOBW for Pareto perturbations with any $\alpha > 1$
- Including results for AwSC regime.
- BOBW for bandits with expert advice

Cost of surrogate probabilities

The surrogate values are not necessarily a probability vector.
$\Longrightarrow$ introduces an additional $\log K$ factor in bandits with expert advice.

Lee, Honda, Ito, Kim, Adaptive learning rates with surrogate probability for follow-the-perturbed-leader, COLT 2026

2-3. The application of surrogate probability (ICML2026)

Broader implication

Surrogate probabilities can replace the use of w_t,
not restricted to learning-rate design.

2-3. Decoupled bandits (ICML2026)

Problem setting

At each round, the learner selects two arms separately.

- An exploitation arm i_t incurs a loss.
- An exploration arm j_t reveals its loss.

Previous FTRL policy (Rouyer and Seldin, 2020)

FTRL with $2/3$-Tsallis entropy

$$i_t \sim w_t \text{ and } j_t \sim g(w_t).$$

- $\mathcal{O}(\sqrt{KT})$ adversarial regret
- $\mathcal{O}(K/\Delta_{\min})$ constant stochastic regret.

Rouyer, Selding, Tsallis-INF for decoupled exploration and exploitation in multi-armed bandits, COLT 2020

2-3. Surrogate exploration probabilities (ICML2026)

Our idea

Replace w_t by computable surrogates in linear time.

- Pareto FTPL with $\alpha = 3$ for exploitation
- Surrogate probabilities for exploration
- No convex optimization or resampling for loss estimation
- Same BOBW guarantee to FTRL with $\mathcal{O}(K)$ time-complexity.

Kim, Lee, Oh, Follow-the-perturbed-leader for decoupled bandits: Best-of-both-worlds and practicality, ICML 2026

2-4. Quick recap

Framework	Design choice	Learning rate	Loss estimator
Recent FTRL	Hybrid regularizer **Main focus**	adaptive to w_t **Main focus**	IW estimator Standard choice
Our FTPL	Pareto perturbation Kept simple	adaptive to surrogate **Main focus**	IW estimator? Standard choice

$$\hat{\ell}_{t,i} = \frac{\mathbb{1}[i_t = i]\ell_{t,i}}{w_{t,i}} \text{ is unbiased IW loss estimator.}$$

Different constructions of the IW estimator

- **FTRL:** w_t is explicitly available and can be used directly.
- **FTPL:** w_{t,i_t}^{-1} must be estimated, usually by resampling methods.

2-4. Resampling as a computational bottleneck

- Selecting an arm with FTPL is computationally simple.
- Geometric resampling (GR) repeatedly runs FTPL until the selected arm appears again $\rightarrow$ estimation of w_{t,i_t}^{-1}.
- Rarely selected arms may require many resampling steps.

Computational cost

Standard GR requires $\mathcal{O}(K^2)$ average computation per round and can be slower than FTRL for large K.

- Indeed, FTRL with Tsallis entropy is faster than FTPL with GR.

Loss estimation becomes the main bottleneck.

2-4. Conditional geometric resampling (ICML2025)

Main idea

Remove perturbation samples that cannot select the currently played arm.
Sample directly from appropriate **conditional** distributions.

- Unbiased estimator of w_{t,i_t}^{-1} with reduced variance
- Applicable to general perturbation distributions
- Same BOBW regret guarantees

$$\underbrace{\mathcal{O}(K^2)}_{\text{standard GR}} \longleftrightarrow \underbrace{\mathcal{O}(K \log K)}_{\text{CGR}}$$

Nearly linear average time-complexity per round

Chen, Lee, Honda, Geometric resampling in nearly linear time for follow-the-perturbed-leader with best-of-both-worlds guarantee in bandit problems, ICML 2025

2-5. A broader role of loss estimation

CGR makes IW estimation computationally efficient for FTPL.

A broader question

Can loss estimation do more than approximate the IW estimator?

- In FTRL, the regularizer controls the behavior of w_t.
- The probability vector w_t determines the adaptive learning rate.
- The loss estimator is usually the standard IW estimator.

For FTPL, the loss estimator itself can become part of the policy design.

2-5. Our current design principle

Framework	Design choice	Learning rate	Loss estimator
Recent FTRL	Hybrid regularizer **Main focus**	Based on w_t **Main focus**	IW estimator Standard choice
Our FTPL	Pareto perturbation Kept simple	Based on surrogates **Main focus**	Feedback-dependent **Main focus**

Our current design principle

Keep the perturbation simple while adapting the learning rate and loss estimator to the problem structure.

2-5. example: weakly observable graph bandits

When arm i_t is played, we observe the losses of its outgoing neighbors.

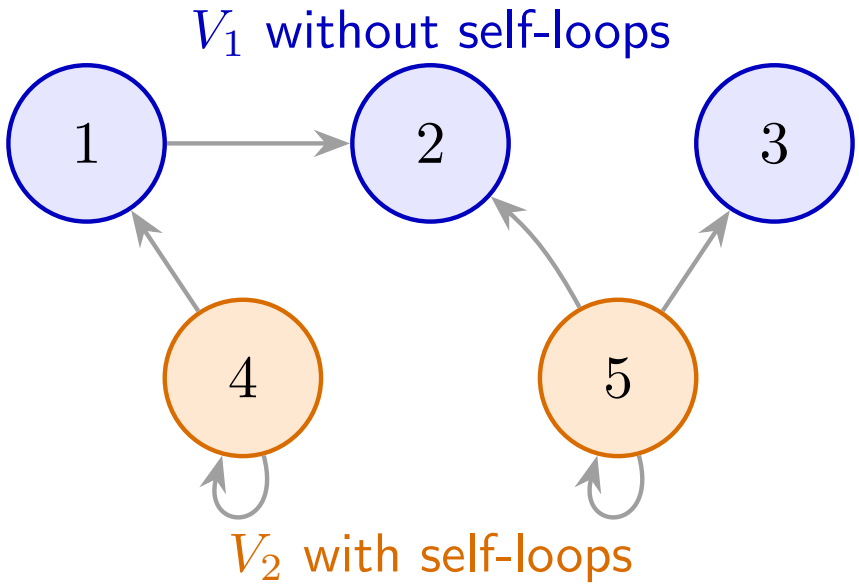
	Feedback structure	**Required treatment**
V_1	No self-loop	Forced exploration
V_2	With a self-loop	MAB-style learning

- Arms in V_1 are observed only by playing their neighbors.
- Arms in V_2 are observed whenever they are selected.

Design implication

The two types of arms require different forms of control.

2-5. example: weakly observable graph bandits



- Arms in V_1 are observed only by playing their neighbors.
- Arms in V_2 are observed whenever they are selected.

Design implication

The two types of arms require different forms of control.

2-5. Arm-dependent regularization in FTRL

Previous FTRL approach (Ito+ 2022, Tsuchiya and Ito 2024)

Use different regularizers according to the feedback structure.

- **Arms in V_1**
  Shannon entropy with forced exploration
- **Arms in V_2**
  $1/2$-Tsallis entropy for MAB-style guarantees

Ito+ (2022) used

$$\psi(x) \approx \sum_{i \in V_1} x \log x + (1-x)\log(1-x) + \sum_{i \in V_2} -2\sqrt{x} - 2\sqrt{1-x}.$$

Ito, Tsuchiya, Honda, Nearly optimal best-of-both-worlds algorithms for online learning with feedback graphs, NeurIPS 2022
Tsuchiya and Ito, A simple and adaptive learning rate for FTRL in online learning with minimax regret of $\Theta(T^{2/3})$ and its application to best-of-both-worlds, NeurIPS 2024

2-5. Arm-dependent regularization in FTRL

Ito+ (2022) used

$$\psi(x) \approx \sum_{i \in V_1} x \log x + (1-x)\log(1-x) + \sum_{i \in V_2} -2\sqrt{x} - 2\sqrt{1-x}.$$

The same graph-based IW estimator is used for both types

$$\hat{\ell}_{t,i} = \frac{\mathbb{1}[i \text{ is observed}]\ell_{t,i}}{P_{t,i}},$$

where $P_{t,i}$ denotes the observation probability.

Different regularizers with a common estimator

Ito, Tsuchiya, Honda, Nearly optimal best-of-both-worlds algorithms for online learning with feedback graphs, NeurIPS 2022

2-5. Arm-dependent loss estimation with FTPL (Preprint)

Common FTPL rule (Lee and Yoo, 2026)

Use the same Pareto perturbation distribution for every arm.

- **Arms in V_1**
  - Use only observations obtained through forced exploration.
  - directly controlled by the exploration distribution.
- **Arms in V_2**
  - Estimate only w_{t,i_t}^{-1} using CGR. $\rightarrow$ **MAB-style loss estimator!**
  - Distribute the estimate across observed arms using graph-dependent weights.
- Adaptive learning rates with surrogates.

Lee, Yoo, Nearly linear-time Follow-the-Perturbed-Leader for weakly observable graph bandits, preprint 2026

2-5. Arm-dependent loss estimation with FTPL (Preprint)

Resulting properties

- Both estimators are unbiased.
- Each estimator utilizes the feedback structure of its arm.
- The policy avoids computing $P_{t,i}$ for every arm.

Common perturbations with arm-dependent estimators

- The first BOBW guarantee of FTPL in difficult problems with near-optimal $\tilde{\mathcal{O}}(T^{2/3})$ adversarial regret.
- $\mathcal{O}(K \log K)$ average time-complexity.

Lee, Yoo, Nearly linear-time Follow-the-Perturbed-Leader for weakly observable graph bandits, preprint 2026

Design lessons for FTPL (personal opinion)

Design choices	Our results	What we learned/noticed
Perturbation	COLT 2024 NeurIPS 2025 Semi-bandit preprint	**A simple Pareto distribution may be enough** for bandit problems.
Learning rate	COLT 2026 ICML 2026 Graph-bandit preprint	**Surrogate quantity can replace w_t.**
Loss estimator	ICML 2025 Graph-bandit preprint	**Loss estimation can exploit the feedback structure efficiently.**

Our current view

For FTPL, it may be more effective to keep the perturbation simple and place the flexibility in adaptive learning rates and problem-dependent loss estimators.

Future directions for surrogate-based design

Probability-dependent loss estimation

Some algorithms use w_t to determine clipping or truncation thresholds in their loss estimators.

- Replace w-dependent thresholds with computable surrogate quantities
- Apply this idea to heavy-tailed bandits
- Extend feedback-dependent estimation to dueling bandits

Beyond FTPL

Surrogate-based updates may also apply to Prod-type methods and other algorithms whose exact action probabilities are expensive to compute.

- In single queuing bandits, Exp3 policy can achieve BOBW results.

(Lee and Kim, preprint 2026)

Beyond FTRL-to-FTPL transfers

A question about research value

How much value remains in
reproducing existing FTRL guarantees through FTPL?

Such a transfer is most compelling when it provides

- a clear computational advantage
- a different algorithmic design principle
- new insights that are specific to randomized policies

Broader research directions

- Learning in games with bandit feedback
- Riemannian game dynamics
- Last-iterate guarantees beyond average regret

The next step is to identify problems where randomized decision rules offer a distinct advantage.